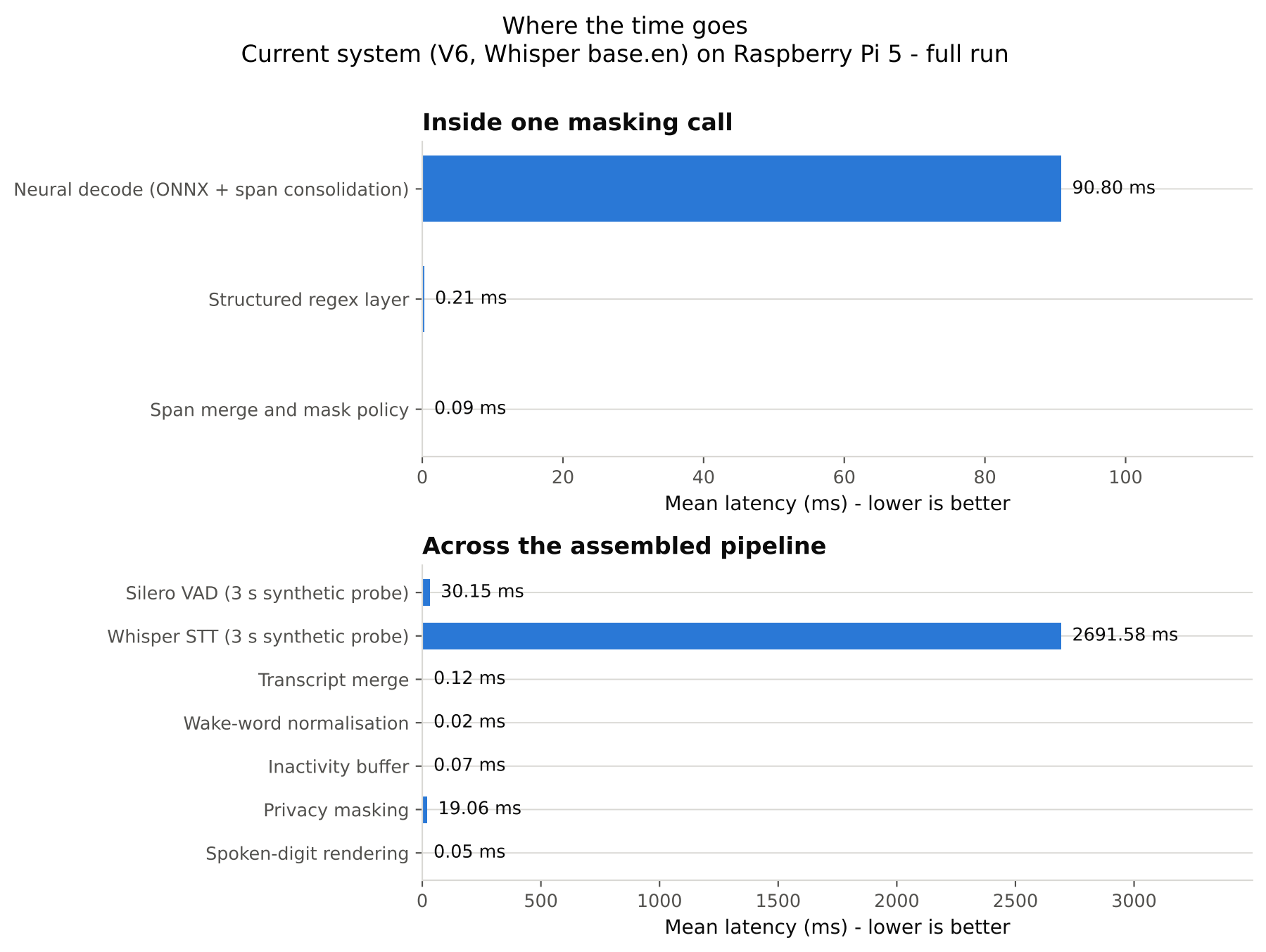
Where the time goes
Current system (V6, Whisper base.en) on Raspberry Pi 5 - full run
Inside one masking call
Neural decode (ONNX + span consolidation)
90.80 ms
Structured regex layer
0.21 ms
Span merge and mask policy
0.09 ms
0
20
40
60
80
100
Mean latency (ms) - lower is better
Across the assembled pipeline
Silero VAD (3 s synthetic probe)
30.15 ms
Whisper STT (3 s synthetic probe)
2691.58 ms
Transcript merge
0.12 ms
Wake-word normalisation
0.02 ms
Inactivity buffer
0.07 ms
Privacy masking
19.06 ms
Spoken-digit rendering
0.05 ms
0
500
1000
1500
2000
2500
3000
Mean latency (ms) - lower is better